

Module 3: Complex Integration

3.1

Functions given by Power Series

Def: A series of the form $\sum_{n=0}^{\infty} a_n z^n$ is called a complex power series or power series.

Def: If $\sum_{n=0}^{\infty} a_n z^n$ converges for $|z| < R$ and diverges for $|z| > R$, then R is called the radius of convergence of the power series and $|z| = R$ is called the Circle of Convergence of the power series.

Taylor's Series

Taylor's Theorem:

Let $f(z)$ be analytic at all points within a circle $|z - z_0| = R$. Then at each point z within $|z - z_0| = R$,

$$f(z) = f(z_0) + (z - z_0) \frac{f'(z_0)}{1!} + (z - z_0)^2 \frac{f''(z_0)}{2!} + \dots + (z - z_0)^n \frac{f^{(n)}(z_0)}{n!} + \dots$$

The expansion in ① on the RHS is called ①

the Taylor's series expansion of $f(z)$ in power of $z - z_0$ or Taylor's series expansion

of $f(z)$ about $z = z_0$.

Examples:

- ① Obtain the Taylor's series expansion of $f(z) = \frac{1}{z}$ about the point $z = 1$ (or, in powers of $(z-1)$).

Sol: The given function $f(z) = \frac{1}{z}$ is analytic at $z = 1$ and not analytic at $z = 0$.
Therefore, the Taylor's series expansion of $f(z) = \frac{1}{z}$ about $z = 1$ is valid in $|z-1| < 1$.

Let $z-1 = w$. Then $z = 1+w$.

$$\begin{aligned}\therefore f(z) = \frac{1}{z} &= \frac{1}{1+w} \\ &= (1+w)^{-1} \\ &= 1 - w + w^2 - w^3 + \dots \text{ for } |w| < 1 \\ &= 1 - (z-1) + (z-1)^2 - (z-1)^3 + \dots \\ &\quad \text{for } |z-1| < 1.\end{aligned}$$

- ② Find the Taylor's series expansion of the function $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ in the region $|z| < 2$.

Sol: Given $f(z) = \frac{z^2-1}{(z+2)(z+3)}$

or, $f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$ (Resolving into partial fractions)

$$= 1 + \frac{3}{2} \frac{1}{\left(1 + \frac{z}{2}\right)} - \frac{8}{3\left(1 + \frac{z}{3}\right)}$$

$$= 1 + \frac{3}{2} \left(1 + \frac{z}{2}\right)^{-1} - \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1}$$

$$= 1 + \frac{3}{2} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots\right)$$

$$- \frac{8}{3} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots\right)$$

$$= 1 + \frac{3}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n} \cdot z^n - \frac{8}{3} \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n} \cdot z^n$$

$$= 1 + \sum_{n=0}^{\infty} (-1)^n \left[\frac{3}{2^{n+1}} - \frac{8}{3^{n+1}} \right] z^n$$

which is the required Taylor's series.

③ Expand $f(z) = \frac{z-1}{z+1}$ in Taylor's series about

(i) $z=0$ and (ii) $z=1$

Sol: $f(z) = \frac{z-1}{z+1}$

$$= \frac{z+1-2}{z+1} = 1 - \frac{2}{z+1}$$

$$= 1 - 2(1+z)^{-1}$$

$$= 1 - 2(1 - z + z^2 - z^3 + z^4 - \dots) \text{ for } |z| < 1$$

$$= -1 + 2(z - z^2 + z^3 - z^4 + \dots) \text{ for } |z| < 1$$

$$= -1 + 2 \sum_{n=1}^{\infty} (-1)^{n-1} \cdot z^n \text{ for } |z| < 1.$$

(ii) Let $z-1=w$. Then $z+1=w+2$.

$$\therefore f(z) = \frac{z-1}{z+1}$$

$$= \frac{w}{(1+w)+1}$$

$$= \frac{w}{2+w} = \frac{w}{2(1+\frac{w}{2})} = \frac{w}{2} (1+\frac{w}{2})^{-1}$$

$$= \frac{w}{2} (1 - \frac{w}{2} + (\frac{w}{2})^2 - (\frac{w}{2})^3 + \dots) \text{ if } |\frac{w}{2}| < 1$$

$$= \frac{w}{2} - (\frac{w}{2})^2 + (\frac{w}{2})^3 - (\frac{w}{2})^4 + \dots \text{ if } |\frac{w}{2}| < 1$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} (\frac{w}{2})^n \text{ if } |w| < 2$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} \left(\frac{z-1}{2}\right)^n \text{ if } |z-1| < 2$$

④ Expand $f(z) = \frac{1}{(z+1)^2}$ in Taylor's series about $z = -i$

Sol: Let $z+i = w$. Then $z = w-i$

$$\therefore f(z) = \frac{1}{(z+1)^2}$$

$$= \frac{1}{((w-i)+1)^2}$$

$$= \frac{1}{[1-i+w]^2}$$

$$= \frac{1}{(1-i)^2 \left(1 + \frac{w}{1-i}\right)^2}$$

$$= \frac{1}{(1-i)^2} \left[1 + \frac{w}{1-i}\right]^{-2}$$

$$= \frac{1}{(1-i)^2} \left[1 - 2\frac{w}{1-i} + 3\left(\frac{w}{1-i}\right)^2 - 4\left(\frac{w}{1-i}\right)^3 + \dots\right]$$

$$= -\frac{1}{2i} \left[1 - 2\left(\frac{w}{1-i}\right) + 3\left(\frac{w}{1-i}\right)^2 - 4\left(\frac{w}{1-i}\right)^3 + \dots\right]$$

$$= \frac{i}{2} \left[1 - \frac{2}{1-i} (z+i) + \frac{3}{(1-i)^2} (z+i)^2 - \frac{4}{(1-i)^3} (z+i)^3 + \dots\right]$$

which is the required expansion.

Or, Given $f(z) = \frac{1}{(z+1)^2}$. So, $f'(z) = \frac{-2}{(z+1)^3}$, $f''(z) = \frac{6}{(z+1)^4}$

and so on $f^{(n)}(z) = (-1)^n \frac{n!}{(z+1)^{n+2}}$

By Taylor's theorem, we have

$$f(z) = f(a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!} (z-a)^n \rightarrow \textcircled{1}$$

To find Taylor's series about $z = -i$, taking

$a = -i$ in $\textcircled{1}$, we get

$$f(z) = f(-i) + \sum_{n=1}^{\infty} \frac{f^{(n)}(-i)}{n!} (z+i)^n$$

$$= \frac{i}{2} + \frac{f'(-i)}{1!} (z+i) + \frac{f''(-i)}{2!} (z+i)^2 + \dots$$

$$= \frac{i}{2} - \frac{2!}{(-i+1)^3 (1!)} (z+i) + \frac{3!}{(-i+1)^4 (2!)} (z+i)^2 - \dots$$

$$= \frac{i}{2} - \frac{2}{(1-i)^2 (1-i)} (z+i) + \frac{3}{(1-i)^2 (1-i)^2} (z+i)^2 - \dots$$

$$= \frac{i}{2} - \frac{2}{-2i(1-i)} (z+i) + \frac{3}{(-2i)(1-i)^2} (z+i)^2 - \dots$$

$$= \frac{i}{2} - \frac{(-i)2}{(-2)(1-i)} (z+i) - \frac{3(-i)}{2(1-i)^2} (z+i)^2 - \dots$$

$$= \frac{i}{2} - \frac{i}{(1-i)} (z+i) + \frac{i}{2} \frac{3}{(1-i)^2} (z+i)^2 - \dots$$

$$\therefore f(z) = \frac{i}{2} \left[1 - \frac{2}{(1-i)} (z+i) + \frac{3}{(1-i)^2} (z+i)^2 - \dots \right]$$